Replacement Section 7 for *Signed weighted residues of the Fricke Hauptmoduln* (compile test)

Audit deliverable

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Context (from the note; for cross-references only)

Theorem 0.1 (Frobenius on the eta quotient; statement unchanged)**.** *Let $\ell \neq p$ be prime. At every supersingular point x of $X_0(\ell)_{\overline{\mathbb{F}}_p}$, $(\Delta(\tau)/\Delta(\ell\tau))^{(p^2-1)/24}(x) = \left(\frac{\ell}{p}\right)$. For $\ell = N \in \mathcal{N}$ this reads $u_N(x)^{(p^2-1)/a_N} = \left(\frac{N}{p}\right)$.*

Lemma 0.2 (Kodaira–Spencer factor of the Fricke isogeny; as in the note)**.** *$B(E',\omega') = N\,B(E,\omega)$, $A(E',\omega') = A(E,\omega)$, $\Delta(N\tau)(x) = N^{-12}\Delta(E',\omega')$, with $\omega = \pi^*\omega'$.*

7 The eta quotient at supersingular points

Throughout this section $\ell \neq p$ is a prime, $M := 576\ell$, $k := (p^2-1)/24$, and

$$D_\ell(\tau) := \frac{\Delta(\tau)}{\Delta(\ell\tau)}, \qquad G(\tau) := \frac{\eta(24\ell\tau)}{\eta(24\tau)} = q^{\ell-1}\prod_{n\geq 1}\frac{1-q^{24\ell n}}{1-q^{24n}} \in q^{\ell-1}\mathbb{Z}[[q]], \qquad G(\tau)^{24} = D_\ell(24\tau)^{-1}.$$

We prove Theorem 0.1: $D_\ell(x)^k = \left(\frac{\ell}{p}\right)$ at every supersingular point x of $X_0(\ell)_{\overline{\mathbb{F}}_p}$. The argument has four steps: (i) the multiplier of G on $\Gamma_0(M)$; (ii) the moduli interpretation of D_ℓ, of G, and of the diamond operators; (iii) the Frobenius action on a lift of x to $X_1(M)$; (iv) the descent.

7.1 The multiplier of G

We use the closed form of the eta multiplier (Rademacher; Petersson; see Knopp, 1970, Ch. 4, §4.1, and Apostol, 1990, Thm. 3.4 for the Dedekind-sum form): for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ with $c > 0$,

$$\eta(\gamma\tau) = \nu(\gamma)\,(c\tau+d)^{1/2}\,\eta(\tau), \qquad \nu(\gamma) = \begin{cases} \left(\frac{d}{c}\right)\, e\Big(\dfrac{(a+d)c - bd(c^2-1) - 3c}{24}\Big), & c \text{ odd}, \\ \left(\frac{c}{d}\right)\, e\Big(\dfrac{(a+d)c - bd(c^2-1) + 3d - 3 - 3cd}{24}\Big), & c \text{ even}, \end{cases} \tag{7.1}$$

where $e(x) = e^{2\pi i x}$, $(c\tau+d)^{1/2}$ is the principal branch, and the symbols are Kronecker symbols (so $\left(\frac{c}{d}\right)$ is defined for $d < 0$, with $\left(\frac{c}{-1}\right) = \operatorname{sgn} c$). We checked (7.1) exactly against the Dedekind-sum formula on 400 random matrices of both parities of c and both signs of d, and the Dedekind-sum formula against 50-digit numerics (file `eta_multiplier_audit.json`).

Lemma 7.1 (multiplier of G)**.** *Let $\chi_\ell(d) := \left(\frac{\ell}{d}\right)$ (Kronecker symbol). For every $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(M)$,*

$$G(\gamma\tau) = \chi_\ell(d)\, G(\tau).$$

The function $d \mapsto \chi_\ell(d)$ on $(\mathbb{Z}/M)^\times$ is the Kronecker character of $\mathbb{Q}(\sqrt{\ell})$: it is a Dirichlet character of conductor ℓ ($\ell \equiv 1 \bmod 4$), 4ℓ ($\ell \equiv 3 \bmod 4$) or 8 ($\ell = 2$), each dividing M; it is even and quadratic, and $\chi_\ell(p) = \left(\frac{\ell}{p}\right)$ is the Legendre symbol.

Proof. If $c = 0$ then $\gamma = \pm \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$, $G(\tau + b) = G(\tau)$ because G has an integral q-expansion, and $\chi_\ell(\pm 1) = 1$. If $c < 0$ replace γ by $-\gamma$, which acts in the same way on $\mathfrak{H}$ and has $\chi_\ell(-d) = \chi_\ell(d)$ since $\left(\frac{\ell}{-1}\right) = 1$. So let $c > 0$. Put $c_1 = c/24$, $c_2 = c/(24\ell)$, $b_1 = 24b$, $b_2 = 24\ell b$ and $\gamma_i = \begin{pmatrix} a & b_i \\ c_i & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. Then $24\gamma\tau = \gamma_1(24\tau)$ and $24\ell\gamma\tau = \gamma_2(24\ell\tau)$, and both automorphy factors equal $(c\tau + d)^{1/2}$, so $G(\gamma\tau) = (\nu(\gamma_2)/\nu(\gamma_1))G(\tau)$. Since $576\ell \mid c$, both c_1 and c_2 are divisible by 24, in particular even, so the second line of (7.1) applies to both. In the quotient of the exponentials every term is a multiple of 24: $(a + d)(c_2 - c_1)$ and $3d(c_2 - c_1)$ because $24 \mid c_i$, and $d(b_2(c_2^2 - 1) - b_1(c_1^2 - 1))$ because $24 \mid b_i$. Hence the exponentials cancel and

$$\frac{\nu(\gamma_2)}{\nu(\gamma_1)} = \left(\frac{c_2}{d}\right)\left(\frac{c_1}{d}\right) = \left(\frac{c_1 c_2}{d}\right) = \left(\frac{\ell c_2^2}{d}\right) = \left(\frac{\ell}{d}\right)\left(\frac{c_2}{d}\right)^2 = \left(\frac{\ell}{d}\right),$$

using $c_1 = \ell c_2$ and $\gcd(c_2, d) = 1$. The statements about χ_ℓ are quadratic reciprocity: for $d > 0$ odd, $\left(\frac{\ell}{d}\right) = \left(\frac{d}{\ell}\right)$ if $\ell \equiv 1 \pmod 4$, $\left(\frac{\ell}{d}\right) = \left(\frac{-1}{d}\right)\left(\frac{d}{\ell}\right)$ if $\ell \equiv 3 \pmod 4$, and $\left(\frac{2}{d}\right) = \chi_8(d)$; in each case this is the Kronecker character of discriminant ℓ, 4ℓ, 8 respectively, which is even because $\mathbb{Q}(\sqrt{\ell})$ is real. For a prime $p \nmid M$, $\chi_\ell(p)$ is the Legendre symbol. □

Remark 7.2. (a) Lemma 7.1 is the case $f = \eta(24\ell\tau)\eta(24\tau)^{-1}$, $N = 576\ell$ of the Gordon–Hughes–Newman theorem (Ono, 2004, Thm. 1.64): $\sum r_\delta = 0$, $\sum \delta r_\delta = 24(\ell - 1)$, $\sum (N/\delta) r_\delta = 24(1 - \ell)$ are $\equiv 0 \pmod{24}$ and $s = \prod \delta^{r_\delta} = \ell$. The proof above is included so that no citation carries the sign. (b) Equivalently, with $y = \eta(\tau)/\eta(\ell\tau)$: if $24\ell \mid c$ and $24 \mid b$ then $y(\gamma\tau) = \left(\frac{\ell}{d}\right) y(\tau)$. The hypothesis $24 \mid b$ cannot be dropped (`eta_multiplier_audit.json`, negative control). (c) Lemma 7.1 was verified exactly for $\ell \le 47$ on random $\gamma \in \Gamma_0(M)$ covering every unit class of d modulo the conductor, with d of both signs, and G was checked to be invariant under $\Gamma_1(M)$.

7.2 Moduli conventions

The μ_M-model. For $M \ge 4$ and over $\mathbb{Z}[1/M]$ let $Y_1(M)$ be the fine moduli scheme of pairs (E, ι) with $\iota : \mu_M \hookrightarrow E$ a closed immersion of group schemes, and $X_1(M) \supset Y_1(M)$ its smooth proper compactification (Deligne and Rapoport, 1973, IV–V); (Gross, 1990, §2); (Katz, 1976, §1). The Tate curve $\mathrm{Tate}(q) = \mathbb{G}_m/q^{\mathbb{Z}}$ over $\mathbb{Z}[1/M]((q))$ with the canonical embedding $\mu_M \subset \mathbb{G}_m$ gives a section ∞ of $X_1(M)$ over $\mathbb{Z}[1/M]$ whose formal completion is $\mathrm{Spf}\, \mathbb{Z}[1/M][[q]]$. The complex points are $\Gamma_1(M)\backslash\mathfrak{H}^*$ via $\tau \mapsto x_\tau := (E_\tau, \iota_\tau)$, $E_\tau = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$, $\iota_\tau(e^{2\pi i/M}) = 1/M$; under $E_\tau \cong \mathbb{C}^\times/q^{\mathbb{Z}}$, $z \mapsto e^{2\pi i z}$, this is the canonical embedding, so the q-expansion of a function at ∞ is its Fourier expansion in $q = e^{2\pi i\tau}$. The diamond operators are the $\mathbb{Z}[1/M]$-automorphisms

$$\langle d\rangle(E, \iota) = (E, \iota \circ [d]), \qquad d \in (\mathbb{Z}/M)^\times.$$

Convention. We write $\langle d\rangle$ for this automorphism of the scheme and $f \circ \langle d\rangle$ for the pull-back of a function; no inverse occurs anywhere below.

Lemma 7.3 (analytic diamond operators)**.** *For* $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(M)$ *one has* $x_{\gamma\tau} = \langle d \rangle x_\tau$. *Consequently, for a function* f *on* $X_1(M)_{\mathbb{C}}$ *with* $f(\gamma\tau) = \chi(d) f(\tau)$ *on* $\Gamma_0(M)$,

$$f \circ \langle d \rangle = \chi(d)\, f.$$

Proof. Multiplication by $c\tau + d$ maps $\mathbb{Z} + \mathbb{Z}\gamma\tau$ onto $\mathbb{Z} + \mathbb{Z}\tau$ (it sends $\gamma\tau \mapsto a\tau + b$ and $1 \mapsto c\tau + d$, and $\det \gamma = 1$), hence is an isomorphism $E_{\gamma\tau} \to E_\tau$. It sends $\iota_{\gamma\tau}(\zeta_M) = 1/M$ to $(c\tau + d)/M \equiv d/M = \iota_\tau(\zeta_M^d)$ because $M \mid c$. So $(E_{\gamma\tau}, \iota_{\gamma\tau}) \cong (E_\tau, \iota_\tau \circ [d])$. Then $(f \circ \langle d \rangle)(x_\tau) = f(x_{\gamma\tau}) = f(\gamma\tau) = \chi(d) f(x_\tau)$. □

Lemma 7.4 (q-expansion principle for functions)**.** *Let* $f \in \overline{\mathbb{Q}}(X_1(M))$ *have Fourier expansion at* ∞ *in* $\mathbb{Q}((q))$. *Then* $f \in \mathbb{Q}(X_1(M))$, *the function field of the* μ_M*-model over* $\mathbb{Q}$.

Proof. For $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, f^σ has expansion $\sigma(\text{coefficients of } f) = f$'s expansion at ∞, because the cusp ∞ and its formal parameter q are defined over $\mathbb{Q}$. So $f^\sigma - f$ vanishes in the completed local ring at ∞, hence identically. □

Remark 7.5 (the point convention). In the model $X_1(M)'$ classifying pairs (E, P) with P of exact order M, the Tate point $(\mathrm{Tate}(q), \zeta_M)$ is only a $\mathbb{Z}[1/M, \zeta_M]$-section (Conrad, n.d., Ex. 2.3); the Galois group acts on the cusps over ∞ through $\sigma_a : (\mathrm{Tate}(q), \zeta_M) \mapsto (\mathrm{Tate}(q), \zeta_M^a)$, i.e. through diamond operators. The same Galois argument then shows that a function with rational q-expansion and $f \circ \langle a \rangle' = \chi(a) f$ satisfies $f^{\sigma_a} = \chi(a)^{-1} f$; so G is defined only over the fixed field of $\ker \chi_\ell$, which is $\mathbb{Q}(\sqrt{\ell})$, and its reduction over $\mathbb{F}_p(\sqrt{\ell}) \subset \mathbb{F}_{p^2}$ (cf. Loeffler, n.d., §3.5). Lemma 7.4 is the statement for the μ_M-model and is what is used below; the point-model statement is recorded only to show that the distinction is real. The two models are isomorphic over $\mathbb{Z}[1/M]$ via $(E, P) \mapsto (E/\langle P \rangle,\ \mu_M \cong E[M]/\langle P \rangle)$ (Weil pairing), but this isomorphism does not commute with the uniformizations by $\mathfrak{H}$, which is why rationality statements may not be transferred between the conventions by changing words only.

The function D_ℓ on $Y_0(\ell)$. For a pair (E, C) over a $\mathbb{Z}[1/\ell]$-scheme, with $\pi : E \to E' = E/C$ the quotient isogeny and ω' any nowhere-vanishing invariant differential on E', put

$$D_\ell(E, C) := \ell^{12} \frac{\Delta(E, \pi^*\omega')}{\Delta(E', \omega')} \in \mathcal{O}(S)^\times. \tag{7.2}$$

This is independent of ω' (both discriminants have weight 12), so it is an invertible function on $Y_0(\ell)_{\mathbb{Z}[1/\ell]}$. On the Tate curve with $C = \mu_\ell$, $E' = \mathrm{Tate}(q^\ell)$ via $t \mapsto t^\ell$, $\pi^*(dt'/t') = \ell\, dt/t$, so $\Delta(E, \pi^*\omega'_{\mathrm{can}}) = \ell^{-12}\Delta(q)$ and $\Delta(E', \omega'_{\mathrm{can}}) = \Delta(q^\ell)$: the q-expansion of (7.2) is $\Delta(q)/\Delta(q^\ell)$. Hence (7.2) is the algebraic incarnation of $\Delta(\tau)/\Delta(\ell\tau)$, with the value at a geometric point $x = (E, C)$ given by (7.2); this is the meaning of "$D_\ell(x)$" in Theorem 0.1 (and $\Delta(\ell\tau)(x) = \ell^{-12}\Delta(E', \omega')$ with $\omega = \pi^*\omega'$, as in Lemma 0.2).

Lemma 7.6 (the degeneracy map)**.** *The rule*

$$\delta(E, \iota) = \Big(E/\iota(\mu_{24}),\quad \iota(\mu_{24\ell})/\iota(\mu_{24}) \Big)$$

defines a morphism $\delta : Y_1(M) \to Y_0(\ell)$ *over* $\mathbb{Z}[1/M]$ *(here* $\mu_{24} \subset \mu_{24\ell} \subset \mu_M$*). On complex points* $\delta(x_\tau) = (E_{24\tau}, \langle 1/\ell \rangle)$, *i.e.* δ *is* $\tau \mapsto 24\tau$, *and*

$$G^{24} = \delta^*(D_\ell^{-1}) \quad \text{in } \mathbb{Q}(X_1(M)).$$

For every $x \in Y_0(\ell)(\overline{\mathbb{F}}_p)$ *there is* $\tilde{x} \in Y_1(M)(\overline{\mathbb{F}}_p)$ *with* $\delta(\tilde{x}) = x$.

Proof. The quotient by the finite étale subgroup $\iota(\mu_{24})$ and the image of $\iota(\mu_{24\ell})$ (cyclic of order ℓ) are defined over $\mathbb{Z}[1/M]$. Analytically, multiplication by 24 induces $E_\tau \to \mathbb{C}/(\mathbb{Z}+\mathbb{Z}\,24\tau) = E_{24\tau}$ with kernel $\langle 1/24\rangle = \iota_\tau(\mu_{24})$, and maps $\iota_\tau(\mu_{24\ell}) = \langle 1/(24\ell)\rangle$ onto $\langle 1/\ell\rangle$. The function D_ℓ at $(E_{\tau'}, \langle 1/\ell\rangle)$ is $\Delta(\tau')/\Delta(\ell\tau')$, so $(\delta^* D_\ell)(x_\tau) = \Delta(24\tau)/\Delta(24\ell\tau) = G(\tau)^{-24}$; two elements of $\overline{\mathbb{Q}}(X_1(M))$ that agree on $\mathbb{C}$-points are equal. For surjectivity on $\overline{\mathbb{F}}_p$-points: given (E, C), choose a cyclic subgroup $K \subset E$ of order 24, put $E_1 = E/K$, so that $E = E_1/K_1$ with $K_1 = E[24]/K$ cyclic of order 24; choose a cyclic subgroup $Z \subset E_1$ of order M with $Z[24] = K_1$ whose image in $E = E_1/K_1$ has ℓ-part C (such Z exist because $E_1[M] \cong (\mathbb{Z}/M)^2$ and $p \nmid M$), and let $\iota : \mu_M \xrightarrow{\sim} Z$ (possible over $\overline{\mathbb{F}}_p$ since μ_M is étale and cyclic). Then $\delta(E_1, \iota) = (E, C)$. □

Proposition 7.7 (descent and integrality of G)**.** *$G \in \mathbb{Q}(X_1(M))$, and G is an invertible regular function on $Y_1(M)_{\mathbb{Z}_{(p)}}$. Its reduction $\overline{G}$ is an invertible regular function on $Y_1(M)_{\mathbb{F}_p}$, defined over $\mathbb{F}_p$, and*

$$\overline{G} \circ \langle d\rangle = \chi_\ell(d)\,\overline{G} \qquad (d \in (\mathbb{Z}/M)^\times), \qquad \overline{G}^{\,24} = \delta^*(\overline{D}_\ell^{-1}).$$

Proof. G is holomorphic and non-vanishing on $\mathfrak{H}$, invariant under $\Gamma_1(M)$ (Lemma 7.1, $\chi_\ell(1) = 1$) and meromorphic at the cusps, so $G \in \mathbb{C}(X_1(M))$; by Lemma 7.4 and the integral q-expansion, $G \in \mathbb{Q}(X_1(M))$. By Lemma 7.6, $F := \delta^*(D_\ell^{-1})$ lies in $\mathcal{O}(Y_1(M)_{\mathbb{Z}[1/M]})^\times$ and $G^{24} = F$. The ring $R := \mathcal{O}(Y_1(M)_{\mathbb{Z}_{(p)}})$ is a normal domain (smooth over $\mathbb{Z}_{(p)}$, flat with integral generic fibre), and $G \in \mathrm{Frac}(R)$ satisfies the monic equation $T^{24} - F = 0$ with $F \in R^\times$; hence $G \in R$, and $G^{-1} = G^{23}F^{-1} \in R$. The identity $G \circ \langle d\rangle = \chi_\ell(d)G$ holds in $\mathbb{C}(X_1(M))$ by Lemmas 7.1 and 7.3, hence in R ($\langle d\rangle$ is an automorphism of $Y_1(M)$ over $\mathbb{Z}[1/M]$), hence after reduction. Reduction of $G^{24} = F$ gives the last identity. □

Remark 7.8. The note's earlier formulation ("a modular unit with unit leading coefficients at all cusps is invertible on $Y_1(M)_{\mathbb{Z}_{(p)}[\zeta_M]}$") is also correct, by the divisor argument on the regular arithmetic surface $X_1(M)_{\mathbb{Z}_{(p)}}$ (irreducible special fibre, cuspidal horizontal divisor), but it needs the expansions at every cusp; the monic-equation argument above needs only ∞ and the moduli description (7.2). The only descent input is Lemma 7.4.

7.3 Frobenius on a lift

Lemma 7.9 (Deuring models)**.** *Let $E_1/\overline{\mathbb{F}}_p$ be supersingular, $p \geq 5$. Then E_1 has a model over $\mathbb{F}_{p^2}$ whose p^2-power Frobenius endomorphism is $[-p]$. For such a model every finite subgroup of order prime to p is $\mathbb{F}_{p^2}$-rational, and E_1/K again has Frobenius $[-p]$.*

Proof. For any model $E/\mathbb{F}_{p^2}$ of a supersingular curve, the p-power Frobenius $F : E \to E^{(p)}$ and the Verschiebung $V : E^{(p)} \to E$ are both purely inseparable of degree p (the Hasse invariant vanishes), so $F^{(p)} : E^{(p)} \to E^{(p^2)} = E$ and V differ by an automorphism u of E: $F^{(p)} = u \circ V$, whence $\pi_{p^2} = F^{(p)} \circ F = u \circ V \circ F = u \circ [p]$. Thus $\pi_{p^2} = p\,u$ with $u \in \mathrm{Aut}(E)$. If $j \neq 0, 1728$ then $u = \pm 1$ and the quadratic twist changes the sign; if $j = 0$ (resp. 1728) the curve $y^2 = x^3 + 1$ (resp. $y^2 = x^3 + x$) over $\mathbb{F}_p$ has trace 0, so $\pi_{p^2} = \pi_p^2 = [-p]$ (Deuring, 1941; Waterhouse, 1969); see also Ono (2026, Lemma 4.3). Since $[-p]$ is central, Galois preserves every subgroup, and the quotient isogeny ϕ satisfies $\pi'\phi = \phi\pi = \phi[-p] = [-p]\phi$, so $\pi' = [-p]$. □

Proof of Theorem 0.1. Let $x = (E, C) \in Y_0(\ell)(\overline{\mathbb{F}}_p)$ be supersingular and choose, by Lemma 7.6, a lift $\tilde{x} = (E_1, \iota) \in Y_1(M)(\overline{\mathbb{F}}_p)$ with $\delta(\tilde{x}) = x$. The curve E_1 is isogenous to E, hence supersingular, so by Lemma 7.9 we may replace (E_1, ι) by an isomorphic pair (same geometric point $\tilde{x}$, same value $\overline{G}(\tilde{x})$) in which E_1 is defined over $\mathbb{F}_{p^2}$ with $\pi_{E_1} = [-p]$. The embedding $\iota : \mu_M \to E_1$ is defined over $\overline{\mathbb{F}}_p$; nothing requires it to be $\mathbb{F}_{p^2}$-rational.

Galois action on the point. Let $\sigma \in \mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_{p^2})$ be $u \mapsto u^{p^2}$. Since $Y_1(M)$ is a fine moduli scheme, σ acts on $Y_1(M)(\overline{\mathbb{F}}_p)$ by $\sigma(E_1, \iota) = (E_1^\sigma, \iota^\sigma) = (E_1, \sigma_{E_1} \circ \iota \circ \sigma_{\mu_M}^{-1})$, where $\sigma_{\mu_M}(\zeta) = \zeta^{p^2}$

and σ_{E_1} is the action on $E_1(\overline{\mathbb{F}}_p)$ through the $\mathbb{F}_{p^2}$-structure, i.e. the Frobenius endomorphism $\pi_{E_1} = [-p]$. Hence

$$\iota^{\sigma}(\zeta) = [-p]\,\iota(\zeta^{p^{-2}}) = [-p][p^{-2}]\,\iota(\zeta) = \iota([-p^{-1}]\zeta), \qquad \text{i.e.} \qquad \sigma(\tilde{x}) = \langle -p^{-1}\rangle \tilde{x},$$

with $-p^{-1}$ read in $(\mathbb{Z}/M)^{\times}$. (In the point convention the same computation gives $\sigma(E_1, P) = (E_1, [-p]P) = \langle -p\rangle'(E_1, P)$; the final answer below is unchanged because χ_ℓ is even and quadratic, but we fix the μ_M-convention and use only the displayed formula.)

Galois action on the value. $\overline{G}$ is defined over $\mathbb{F}_p$ (Proposition 7.7), so $\overline{G}(\sigma\tilde{x}) = \sigma(\overline{G}(\tilde{x})) = \overline{G}(\tilde{x})^{p^2}$. Combining with the diamond identity of Proposition 7.7,

$$\overline{G}(\tilde{x})^{p^2} = \overline{G}(\langle -p^{-1}\rangle \tilde{x}) = \chi_\ell(-p^{-1})\,\overline{G}(\tilde{x}) = \chi_\ell(p)\,\overline{G}(\tilde{x}) = \left(\frac{\ell}{p}\right)\overline{G}(\tilde{x}),$$

because $\chi_\ell(-1) = 1$ and $\chi_\ell(p^{-1}) = \chi_\ell(p)^{-1} = \chi_\ell(p)$. Since $\overline{G}$ is invertible on $Y_1(M)_{\mathbb{F}_p}$, $\overline{G}(\tilde{x}) \neq 0$ and

$$\overline{G}(\tilde{x})^{p^2-1} = \left(\frac{\ell}{p}\right). \tag{7.3}$$

No coefficient-field factor enters: the only Galois action used is that of σ on the point $\tilde{x}$, and $\overline{G}$ is σ-invariant as a function.

Descent to $X_0(\ell)$. By Proposition 7.7, $\overline{G}(\tilde{x})^{24} = \overline{D}_\ell(\delta\tilde{x})^{-1} = D_\ell(x)^{-1}$. Therefore

$$D_\ell(x)^k = \overline{G}(\tilde{x})^{-24k} = \overline{G}(\tilde{x})^{-(p^2-1)} = \left(\frac{\ell}{p}\right)^{-1} = \left(\frac{\ell}{p}\right).$$

This holds at every geometric supersingular point, including those over $j = 0$ and $j = 1728$: the argument takes place on the fine moduli scheme $Y_1(M)$, and $D_\ell(x)$ is defined by (7.2) independently of automorphisms of (E, C). For $\ell = N \in \mathcal{N}$, $D_N = u_N^{N-1}$ and $(N-1)k = (p^2-1)/a_N$. □

Remark 7.10 (comparison with Ono). Ono (2026, Thm. 1.2) (v2, 3 July 2026) proves $R_\ell^{(p^2-1)/12} = 1$ at supersingular points for $R_\ell = \Delta(\ell\tau)/\Delta(\tau) = D_\ell^{-1}$, via $R_\ell(\tilde{P})^{(p^2-1)/12} = \kappa(E')/\kappa(E)$ (his Lemma 4.4) and $\kappa(E) = \Delta(E, \omega)^{(p^2-1)/12} = (-1)^{(p+1)/2}$ on Deuring models with $\mathbb{F}_{p^2}$-rational ω (his Thm. 1.3). His Remark 4 observes that the $(p^2-1)/24$ power can be -1 ($p = 11$, $\ell = 2$: $R_2 \in \{6, 7, 2\}$, fifth powers -1), that the values of $g_\ell = \eta(\ell z)/\eta(z)$ need not lie in $\mathbb{F}_{p^2}$ (they generate $\mathbb{F}_{11^4}$ there), and that a proof descending $\Delta^{1/24}$ or $\Delta^{1/12}$ to $\mathbb{F}_{p^2}$ must fail. Theorem 0.1 determines the sign: $R_\ell^{(p^2-1)/24} = \left(\frac{\ell}{p}\right)$, equivalently $R_\ell(\tilde{P})$ is a 24-th power in $\mathbb{F}_{p^2}^{\times}$ iff $\left(\frac{\ell}{p}\right) = 1$. This is consistent with Remark 4: $\left(\frac{2}{11}\right) = -1$, and (7.3) shows $\overline{G}(\tilde{x}) \in \mathbb{F}_{p^4} \setminus \mathbb{F}_{p^2}$ exactly when $\left(\frac{\ell}{p}\right) = -1$. The proof above does not descend any root of Δ to $\mathbb{F}_{p^2}$: it uses that $\overline{G}$ is an $\mathbb{F}_p$-rational *function* on $Y_1(576\ell)$, and tracks its values through the diamond character. The statement $R_\ell^{(p^2-1)/12} = 1$ for curves isogenous over $\mathbb{F}_{p^2}$ is already in Baker (1998, §2), as a consequence of the Gross–Landweber congruence.

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